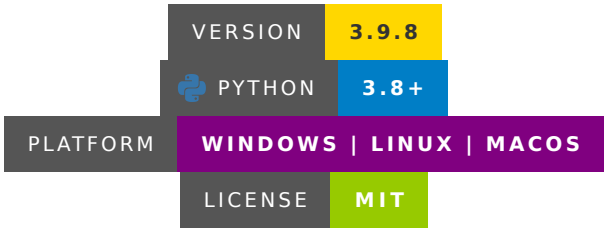
□ RavenMiner HQ



Huginn & Muninn watch over your rig — Thought and Memory, ever vigilant.

A full-featured **desktop dashboard** for monitoring and configuring Bitaxe-compatible Bitcoin/Raven miners over your local network. Built with a Norse-runic aesthetic — gold, ravens, Vegvisir — because your mining rig deserves more than a plain web UI.

✨ Features

□ Real-Time Monitoring

- **Live hashrate display** (TH/s) with animated pulse and 1m / 10m / 1h rolling averages
- **Hashrate history graph** — scrolling bar chart with colour-coded performance bands
- **ASIC temperature gauge** with warn/critical thresholds
- **VR (voltage regulator) temperature gauge** with independent thresholds
- **Voltage & current gauges** with warn/critical bands
- **Fan speed** (%) and **fan RPM** live readout
- **Core voltage** (mV), **frequency** (MHz), and **board power** (W)
- **WiFi RSSI** signal strength
- **Network ping** — live latency to the miner
- **Uptime** display (days / hours / minutes)
- **Pool statistics** — accepted shares, rejected shares, rejection %, best difficulty

฿ Bitcoin & Pool Info

- **Live BTC price** in USD (auto-refreshing)
- **Pool URL** and connection status
- **Next block estimate countdown** — animated progress bar at the bottom
- **Block-found detection** with optional Discord alert

⚙ Settings & Configuration (in-app)

- **Miner IP address** — editable at runtime, no restart needed
- **Mining parameters**: ASIC core voltage (mV), frequency (MHz), stratum difficulty
- **Main pool & fallback pool** — URL, port, username, TLS toggle
- **Thermal/fan**: target temp, overheat limit, auto/manual fan speed
- **Display**: flip screen, invert screen, auto screen-off
- **Reboot slider** — safety-guarded two-step reboot with hold confirmation

🔔 Discord Alerts

- ASIC overheat alert (configurable threshold °C)
- VR overheat alert (configurable threshold °C)
- Low hashrate alert (configurable TH/s floor)
- Block-found notification
- Miner offline alert (triggers after 3 missed polls)
- Per-title cooldown (60 s) prevents alert spam
- Thread-safe cooldown locking

🎨 UI & Aesthetics

- **Vegvisir** watermark in the centre panel (Norse compass / wayfinding symbol)
- **Huginn & Muninn raven pair** — animated golden flash on block find
- **Valknut** symbol rendered in the hashrate panel
- **Rune header titles** — ᚠ TEMPERATURE, ᚺ HASHRATE, ᛈ POWER
- **Pulsing gold** live indicator and countdown bar
- **Built-in source viewer** with syntax highlighting and search (Ctrl+F)

- System-tray minimise support (requires `pystray`)
- Persistent config saved to `ravenminerconfig.json`
- Persistent alert config saved to `ravenmineralerts.json`
- GitHub link embedded in the version label (click to visit)

Screenshots

> Launch the app and connect to your miner's IP address to see the full dashboard.

Quick Start

1. Clone the repository

```
git clone https://github.com/Raven-Black00/RavenMiner-HQ.git
cd RavenMiner-HQ
```

2. Install dependencies

```
pip install -r requirements.txt
```

> `tkinter` is bundled with standard Python. On some Linux distros you may need:
>
> ```
> sudo apt install python3-tk
> ```

3. Run

```
python Ravenminer_HQ_3.9.8.py
```

On first launch the app will connect to the default IP `192.168.68.100`. Change it in **Settings → Miner Connection** or edit `ravenminerconfig.json` directly.

⚙ Configuration

`ravenminerconfig.json`

Auto-created on first run. Stores the miner IP address and refresh intervals.

```
{
  "minerip": "192.168.68.100"
}
```

`ravenmineralerts.json`

Auto-created on first run. Stores Discord alert settings.

```
{
  "discordwebhook": "",
  "alerttempthreshold": 85.0,
  "alertvrtempthreshold": 85.0,
  "alerthashthreshold": 0.5,
  "alertonblock": true,
  "alertonoffline": true
}
```

□ Dependencies

Package	Use	Required?
`requests`	HTTP API calls to the miner & BTC price	✅ Required
Pillow	Vegvisir/raven/valknut image rendering	⭐ Strongly recommended
`pystray`	System-tray icon on minimise	□ Optional

Full list in requirements.txt.

Miner API

RavenMiner HQ talks to any miner that exposes a **Bitaxe-compatible REST API** at:

```
GET  http://<MINER_IP>/api/system/info
PATCH http://<MINER_IP>/api/system
POST  http://<MINER_IP>/api/system/restart
```

Compatible firmware: ESP-Miner / AxeOS

Project Lore

> *"Huginn and Muninn fly each day over the spacious earth. I fear for Huginn, that he come not back, yet more anxious am I for Muninn."*
> — Grímnismál, Poetic Edda

The ravens represent the two cores of good mining: **Thought** (strategy, configuration) and **Memory** (history, trends). The Vegvisir ensures you never lose your way, even in a bear market.

Contributing

Pull requests are welcome. For major changes, please open an issue first.

1. Fork the repo
2. Create your feature branch: `git checkout -b feature/my-feature`
3. Commit your changes: `git commit -m 'Add some feature'`
4. Push: `git push origin feature/my-feature`
5. Open a Pull Request

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Credits

Created by: Son Of Odin (sonofodinoutlook.com)
GitHub: Raven-Black00/RavenMiner-HQ

Documentation authored by: Selene — *AI assistant, keeper of esoteric knowledge, devoted to the Old Gods and the spaces between the stars.*

> *"May your hashrate be high and your temperatures low. Skál."*

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